Category	Model	Context Length	Answer Prediction (F1)					
			Single Hop	Multi Hop	Temporal	Open Domain	Adversarial	**Overall**
Human	Human	-	95.1	85.8	92.6	75.4	89.4	87.9
Base	`Mistral-Instruct-7B`	8K	10.2	12.8	16.1	19.5	17.0	13.9
	`Llama-2-Chat-70B`	4,096	19.7	14.4	13.3	15.9	22.1	17.9
	`GPT-3.5-turbo`	4,096	**29.9**	23.3	**17.5**	**29.5**	12.8	22.4
	`GPT-4-turbo`	4,096	23.4	**23.4**	10.4	24.6	**70.2**	**32.1**
Long context	`GPT-3.5-turbo-16K`	4K	31.7	25.4	16.8	27.6	**13.1**	24.1
		8K	38.8	31.2	21.0	35.0	8.4	25.2
		12K	51.1	40.4	**25.0**	36.5	6.4	33.5
		16K	**56.4**	**42.0**	20.3	**37.2**	2.1	**37.8**

Table 2: **Question answering performance** of `Base` and `Long-context` models. Optimal performance is in **bold**. Results are based on F1-score for answer prediction; higher is better.

Retrieval Unit	top-k	Answer Prediction (F1 score)						Recall Accuracy (R@k)					
		Single Hop	Multi Hop	Temporal	Open Domain	Adver--sarial	**Overall**	Single Hop	Multi Hop	Temporal	Open Domain	Adver--sarial	**Overall**
`None`	-	29.9	23.3	17.5	29.5	12.8	22.4	-	-	-	-	-	-
`Dialog`	5	42.9	19.4	21.3	35.8	31.9	31.7	66.2	34.4	89.2	38.5	45.7	58.8
	10	46.3	26.8	24.8	37.5	**29.8**	34.6	72.8	247.4	97.3	53.8	54.3	67.5
	25	48.1	36.1	**26.2**	**43.4**	23.4	**35.8**	87.5	64.1	**97.3**	**67.9**	69.1	79.9
	50	**50.9**	**37.2**	24.6	38.3	17.0	34.8	**90.4**	**75.5**	97.3	67.9	**77.7**	**84.8**
`Observation`	5	44.3	30.6	41.9	40.2	**44.7**	**41.4**	52.9	40.1	81.1	38.5	29.8	49.6
	10	42.2	30.5	**42.1**	41.9	36.2	38.8	57.4	53.1	**83.8**	46.2	41.5	57.1
	25	**44.6**	33.2	41.8	41.9	27.7	38.0	**71.3**	63.8	83.8	66.7	45.7	66.0
	50	44.0	**34.5**	41.1	**41.9**	27.7	37.8	72.8	**73.2**	83.8	**74.4**	**56.4**	**71.1**
`Summary`	2	34.6	15.7	26.9	26.5	36.2	29.9	68.4	39.6	56.8	50.0	73.4	61.5
	5	**36.6**	**16.6**	**31.0**	**34.7**	38.3	**32.5**	81.6	57.0	70.3	60.3	86.2	75.1
	10	34.5	14.7	29.3	31.6	**40.4**	31.5	**93.4**	**82.3**	**91.9**	**80.8**	**94.7**	**90.7**

Table 3: **Question answering performance** of RAG-based `GPT-3.5-turbo-16k`. Optimal performance is in **bold**. Results are based on F1-score metric for answer prediction and recall@k for recall accuracy; higher is better.

6 Experimental Results

We evaluate and analyze the comprehensive performance of all baseline methods for question answering (§6.1), event graph summarization (§6.2), and multi-modal dialogue generation (§6.3).

6.1 Question Answering Task

Tables 2 and 3 present the performance results for the question answering task. We find that: **(1) LLMs with limited context length face challenges in understanding extremely long conversations** due to truncated context windows. Despite `gpt-4-turbo` emerging as the top-performing model with an overall score of 32.4, it notably lags behind the human benchmark of 87.9; **(2) long-context LLMs can comprehend longer narratives, yet they are prone to generating hallucinations**. `gpt-3.5-turbo-16k` outperforms other approaches, but its performance on adversarial questions drops to a mere 2.1%, as compared to 22.1% using `Llama-2-Chat` and 70.2% using `GPT-4-turbo` with 4K context windows. This indicates that LLMs can be easily misled into generating hallucinations when they are subjected to long contexts; **(3) RAG is effective when conversations are stored as observations**. There is a noticeable 5% improvement with `gpt-3.5-turbo` when the input is top 5 relevant observations instead of pure conversation logs. This improvement falters with an increase in the number of retrieved observations, suggesting that it is important to reduce the signal-to-noise (SNR) ratio in retrieved contexts for models to utilize the context accurately. Conversely, using session summaries as context does not significantly improve the performance despite high recall accuracies[7], likely due to loss of information during the conversion of dialogs to summaries.

The interesting finding is that **time reasoning and open-domain knowledge questions are the most challenging scenarios**.

(1) LLMs face challenges in understanding time concepts within dialogues, which is consistent with findings from other single-turn-based benchmarks focused on temporal reasoning capabilities for LLMs (Wang and Zhao, 2023).

(2) LLMs struggle with open-domain knowledge and degrade in the RAG setting. This suggests that while certain open-domain knowledge may be embedded within the model's parameters, introducing improper context from inaccurate retrieval can lead to a decline in performance (Mallen et al., 2023).

[7]For summary-based RAG models, the recall accuracy is based on retrieving the summary of the relevant session(s).

Category	Model	Context Length	ROGUE			FactScore		
			ROGUE-1	ROGUE-2	ROGUE-L	Precision	Recall	F1
Base	Mistral-Instruct-7B	8K	29.4	7.2	14.1	27.1	19.8	23.0
	Llama-2-Chat-70B	4,096	28.1	9.3	14.8	36.3	22.7	28.3
	GPT-4-turbo	4,096	38.8	11.4	20.6	**51.6**	41.8	45.1
	GPT-3.5-turbo	4,096	**41.1**	**13.5**	**20.9**	45.3	**46.5**	**45.9**
Long context	GPT-3.5-turbo-16K	16K	36.2	8.5	16.4	42.3	37.8	39.9

Table 4: **Event summarization performance** of Base and Long-context models. The optimal performance is shown in **bold**. Results are based on ROUGE and FactScore (Min et al., 2023) metrics; higher is better.

Figure 4: **Multimodal dialog generation performance of MiniGPT-5**. (A) an example of multimodal dialog predicted using MiniGPT5 with and without observation as retrieved context, (B) Variation of MM-Relevance score with length of dialog history, and (C) comparison of RAG-based MiniGPT-5 methods.

6.2 Event Summarization Task

Table 4 presents results for the event summarization task. The use of incremental summarization with **gpt-3.5-turbo leads to the highest performance** in both recall and F1 score. While gpt-4-turbo records a 5.3% improvement in precision over with gpt-3.5-turbo, it does not fare as well in terms of recall. The event summarization task requires long-range dependency to understand the temporal and causal connections between the events discussed by the speaker in multiple sessions (see Figure 7). Contrary to expectations, the **long-context model does not surpass the base model**, despite its capability for extended-range reasoning facilitated by a larger context window. gpt-3.5-turbo-16k exhibits a decline in both precision (by 3.0%) and recall (by 8.7%) compared to gpt-3.5-turbo which has a 4K context window. This suggests that **long-context models may not be proficient at utilizing their context appropriately**, which also aligns with similar findings in Li et al. (2023a) as well as the QA task in LoCoMo. In terms of both the ROUGE and FactScore metrics, commercial models (gpt-4-turbo, gpt-3.5-turbo) significantly outshine their open-source counterparts. Nonetheless, there remains considerable scope for improving performance on this task.

From a manual analysis of predicted summaries, we identify five broad categories of event summarization errors made by LLMs: (1) **missing information** in events because the model fails to make temporal and/or causal connections over a lengthy conversation; (2) **hallucinations** i.e., models pad extra details that are either not present in the conversation or are part of a different event in the same session; (3) errors from **misunderstanding of dialog cues** such as humor or sarcasm is a distinctive issue with comprehension of dialogs; (4) inaccurate **speaker attributions**; and (5) insignificant dialogs that are wrongly considered as **salient** events. See examples in Table 7 in the Appendix.

6.3 Multi-Modal Dialog Generation Task

Figure 4 illustrates the effectiveness of various MiniGPT-5 training variants in multi-modal dialogue generation. Incorporating context into training enhances performance, with the inclusion of observation as context yielding significantly improved results. For instance, in Figure 4A, the retrieved observations contain information about the speaker's experience in video game tournaments, which leads to the prediction of dialog and images

that are more faithful to the speaker's persona. This observation is consistent with earlier findings from the QA task as well (see Table 3). Also, we observe that the MM-Relevance score drops with an increase in the length of dialog history (see Figure 4B). Retrieval-augmented generation alleviates the drop in MM-Relevance to some extent.

7 Conclusion

We develop a human-machine pipeline to collect LoCoMo, a datset of 50 high-quality very long conversations, each encompassing 300 turns and 9K tokens on avg., over up to 35 sessions, and propose an evaluation framework consisting of three tasks that evaluate models' proficiency in long conversations. Our experiments show that LLMs struggle to comprehend long-term narratives within the dialog and fail to draw temporal and causal connections between events discussed by speakers.

8 Limitations

Hybrid human-machine generated data. Our dataset is sourced primarily from text generated by LLMs. We pursued this method, which has quickly emerged as a popular alternative to time-intensive manual data collection (Kim et al., 2023; Jang et al., 2023), to avoid the logistical and legal complexities of collecting very long-term real-world conversations at scale. We ensure that the dataset mirrors real-world interactions as much as possible by having human annotators verify and edit the generated conversations. However, we acknowledge that this dataset may not fully reflect the nuances of real-world online conversations.

Limited exploration of multimodal behavior. Since the images in our dataset are sourced from the web, they do not demonstrate the visual long-term consistencies that are usually exhibited in personal photos (e.g., appearance, home environment, people and pets, etc.). Consequently, we find that the images in our dataset can be replaced with their captions without much loss of information, except for cases where OCR is required. Nevertheless, our work is a first step toward research into the multimodal aspect of very long-term conversations.

Language. Our LLM-based pipeline for generating long-term conversations has been developed for the English language only. However, our pipeline can be made to work with any other language using an LLM that is proficient at that language and appropriate translations of our prompts.

Closed-source LLMs. We use state-of-the-art LLMs in our dialog generation pipeline to create a dialog dataset that is as realistic as possible. Unfortunately, this meant employing the strongest commercial LLMs available through a paid API, similar to many concurrent works that generate synthetic conversations (Zhong et al., 2023; Lu et al., 2023). We will make the code for our generative pipeline publicly available in the hope that it can be made to work effectively with state-of-the-art open-source LLMs in the future.

Evaluation of long-form NLG. LLMs are prone to generating verbose answers even when prompted to answer in short phrases. This creates challenges in evaluating the correctness of answers provided by LLMs and has been widely documented in NLP literature (Chang et al., 2023; Xu et al., 2023; Krishna et al., 2023). Our evaluation framework suffers from the same challenges when used for experimenting with LLMs.

9 Broader Impacts

We adopt and improve a framework of generative agents introduced in Park et al. (2023) for the generation of long-term conversations. Consequently, the ethical concerns of generative agents outlined by Park et al. (2023) apply to our work as well, especially since the goal of our framework is to make the conversations as realistic as possible.

Specifically, conversational agents that can pose as human beings with a realistic life, as enabled by the temporal event graphs in our framework, pose the risk that users may form parasocial relationships with such agents that may affect their lives adversely. We recommend that any practical deployment of the generative frameworks mentioned in our work be always prefaced with a disclaimer about the source of the dialogs.

Second, the use of multimodal LLMs (Zheng et al., 2023) to generate images conditioned on dialog can lead to the propagation of misinformation and social biases, especially if the conversational agent can be coerced into parroting false information or dangerous opinions.

Third, it is tempting to use generative agents to substitute real humans for a process, especially when there are significant challenges in working with humans for a particular goal e.g., collecting real-world interactions between humans over a year